

eitkit — Electrical Impedance Tomography Reconstruction Toolkit

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1 Overview

Electrical Impedance Tomography (EIT) reconstructs the interior conductivity distribution $\sigma(x)$ of a domain $\Omega \subset \mathbb{R}^2$ from boundary voltage measurements. The mathematical problem, posed by Calderón [1], is severely ill-posed: small perturbations in boundary data can induce arbitrarily large errors in any reconstruction [2].

eitkit is an open-source Python package that implements the complete **difference EIT** pipeline:

Stage	Implementation
Mesh generation	DistMesh2D triangular mesh on the unit disc
Forward problem	P1-FEM with gap electrode model
Jacobian	Adjoint method with sparse LU factorisation reuse
Inverse problem	Tikhonov (L^2) + L-curve and TV (L^1) via ADMM

The toolkit requires only NumPy, SciPy, and Matplotlib. Version 0.1.1 provides a demo notebook, five synthetic phantom shapes, and 154 unit tests. The source code, installation instructions, and interactive examples are available at

<https://github.com/abdgaftunde/eit-reconstruction-toolkit>

2 Forward Problem

2.1 Governing Equation

Let $\Omega \subset \mathbb{R}^2$ be a bounded Lipschitz domain. Given a conductivity $\sigma \in L^\infty(\Omega)$ with $\sigma(x) \geq c > 0$, the electric potential $u \in H^1(\Omega)$ satisfies

$$-\nabla \cdot (\sigma \nabla u) = 0 \quad \text{in } \Omega, \tag{1}$$

$$\sigma \frac{\partial u}{\partial n} = j \quad \text{on } \partial\Omega, \tag{2}$$

where j is the applied current density with $\int_{\partial\Omega} j \, ds = 0$. A ground condition $u(x_0) = 0$ at one interior node ensures uniqueness.

2.2 Gap Electrode Model

Each of the L electrodes is modelled as a single boundary node. A stimulation pair (e^+, e^-) injects current $+I$ at electrode e^+ and extracts $-I$ at e^- . The Neumann load vector $f \in \mathbb{R}^N$ has entries

$$f_i = \begin{cases} +I & \text{node } i \text{ is electrode } e^+, \\ -I & \text{node } i \text{ is electrode } e^-, \\ 0 & \text{otherwise.} \end{cases}$$

2.3 P1 Finite Element Discretisation

The domain is triangulated with M linear elements on N nodes. Writing $u = \sum_{i=1}^N u_i \varphi_i$ with P1 hat functions φ_i , the weak form yields

$$K(\sigma) u = f, \quad (3)$$

where the global stiffness matrix is assembled element-wise:

$$K = \sum_{e=1}^M K^e, \quad K_{ij}^e = \frac{\sigma_e}{4A_e} (b_i b_j + c_i c_j), \quad i, j \in \{0, 1, 2\}.$$

The gradient coefficients are $b_i = y_{i+1} - y_{i+2}$, $c_i = x_{i+2} - x_{i+1}$ (indices mod 3), and A_e is the element area. Assembly is vectorised over elements using NumPy advanced indexing, producing a `csr_array` for efficient sparse arithmetic.

2.4 Measurement Protocol

The standard **adjacent drive / adjacent measurement** protocol is used. For L electrodes there are L drive steps; step k sources current at electrode k and sinks at $(k+1) \bmod L$. Per drive step, $L-3$ voltage measurements are taken across adjacent electrode pairs $(k+2, k+3), \dots, (k+L-2, k+L-1)$ (all indices modulo L). The total number of measurements is $P = L(L-3)$; for $L = 16$ this gives $P = 208$.

2.5 Difference EIT and the Jacobian

Absolute voltages are difficult to calibrate experimentally. Difference EIT measures voltage changes relative to a reference state σ_0 : $\delta V = V(\sigma) - V(\sigma_0)$. Linearising around σ_0 gives the Born approximation

$$\delta V \approx J(\sigma_0) \delta \sigma, \quad (4)$$

where $J \in \mathbb{R}^{P \times M}$ is the Jacobian and $\delta \sigma = \sigma - \sigma_0 \in \mathbb{R}^M$ is the conductivity perturbation.

The Jacobian is computed by the **adjoint method**. For measurement i associated with drive step k and electrode pair (e^+, e^-) ,

$$J_{ie} = - \int_{\Omega_e} \nabla u_k \cdot \nabla \phi_i d\Omega, \quad (5)$$

where u_k is the forward potential for drive step k and ϕ_i is the adjoint potential (unit current injected at e^+ , extracted at e^-). For the adjacent protocol, at most L adjoint solves are needed, and the factorised stiffness matrix is reused for all solves via `scipy.sparse.linalg.splu`.

3 Inverse Problem

3.1 Tikhonov Regularisation (L^2)

The Tikhonov solution minimises

$$\delta\hat{\sigma} = \arg \min_{\delta\sigma} \left\{ \frac{1}{2} \|J\delta\sigma - \delta V\|_2^2 + \lambda \|\delta\sigma\|_2^2 \right\}, \quad (6)$$

with closed-form solution $\delta\hat{\sigma} = (J^T J + \lambda I)^{-1} J^T \delta V$. The regularisation parameter λ is chosen automatically via the L-curve method [4]: the optimal λ corresponds to the point of maximum curvature on the log-log plot of residual norm against solution norm.

3.2 Total Variation (L^1 / ADMM)

TV regularisation promotes piecewise-constant reconstructions:

$$\delta\hat{\sigma} = \arg \min_{\delta\sigma} \left\{ \frac{1}{2} \|J\delta\sigma - \delta V\|_2^2 + \alpha \|D\delta\sigma\|_1 \right\}, \quad (7)$$

where $D \in \mathbb{R}^{F \times M}$ is a sparse finite-difference operator on element-to-element edges. The problem is solved by the Alternating Direction Method of Multipliers (ADMM) [5] with splitting $z = D\delta\sigma$. The σ -update is a quadratic system solved by direct factorisation, the z -update is element-wise soft-thresholding, and the penalty parameter ρ is auto-scaled as $\rho = \text{tr}(J^T J) / \text{tr}(D^T D)$.

4 Package Structure

```
eitkit/
  mesh/           DistMesh2D generation, electrode placement
  forward/        P1 FEM assembly, gap model, adjoint Jacobian
  protocol/       Adjacent drive/meas patterns, AWGN noise
  inverse/
    classical/    Tikhonov + L-curve, TV-ADMM
  utils/          Phantom shapes (5), publication-quality plots
```

The core dependency stack is NumPy ≥ 1.24 , SciPy ≥ 1.10 , and Matplotlib ≥ 3.7 . Optional extras include PyTorch and JAX for future learning-based extensions.

5 References

References

- [1] A.P. Calderón. On an inverse boundary value problem. In *Seminar on Numerical Analysis and its Applications to Continuum Physics*, Rio de Janeiro, 1980.
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- [3] P.-O. Persson and G. Strang. A simple mesh generator in MATLAB. *SIAM Review*, 46(2):329–345, 2004.
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- [5] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein. Distributed optimization and statistical learning via the alternating direction method of multipliers. *Foundations and Trends in Machine Learning*, 3(1):1–122, 2011.
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